

Qno3. Pseudocode.

DECLARE Def : STRING

DECLARE Sum, marks, Average : Real

DECLARE N, student : INTEGER

INPUT N

// No. of students

FOR student = 1 to N

Sum = 0

FOR subj = 1 to 5

INPUT marks

Sum = Sum + marks

IF mark < 33

THEN

Def = True

END IF

END FOR

Average = Sum / N

IF Def == "True"

THEN

Print ("Fail Subject deficiency")

ELSE IF Average >= 80

THEN

Print ("Distinction")

ELSE IF Average >= 60

THEN

Print ("Pass")

ELSE

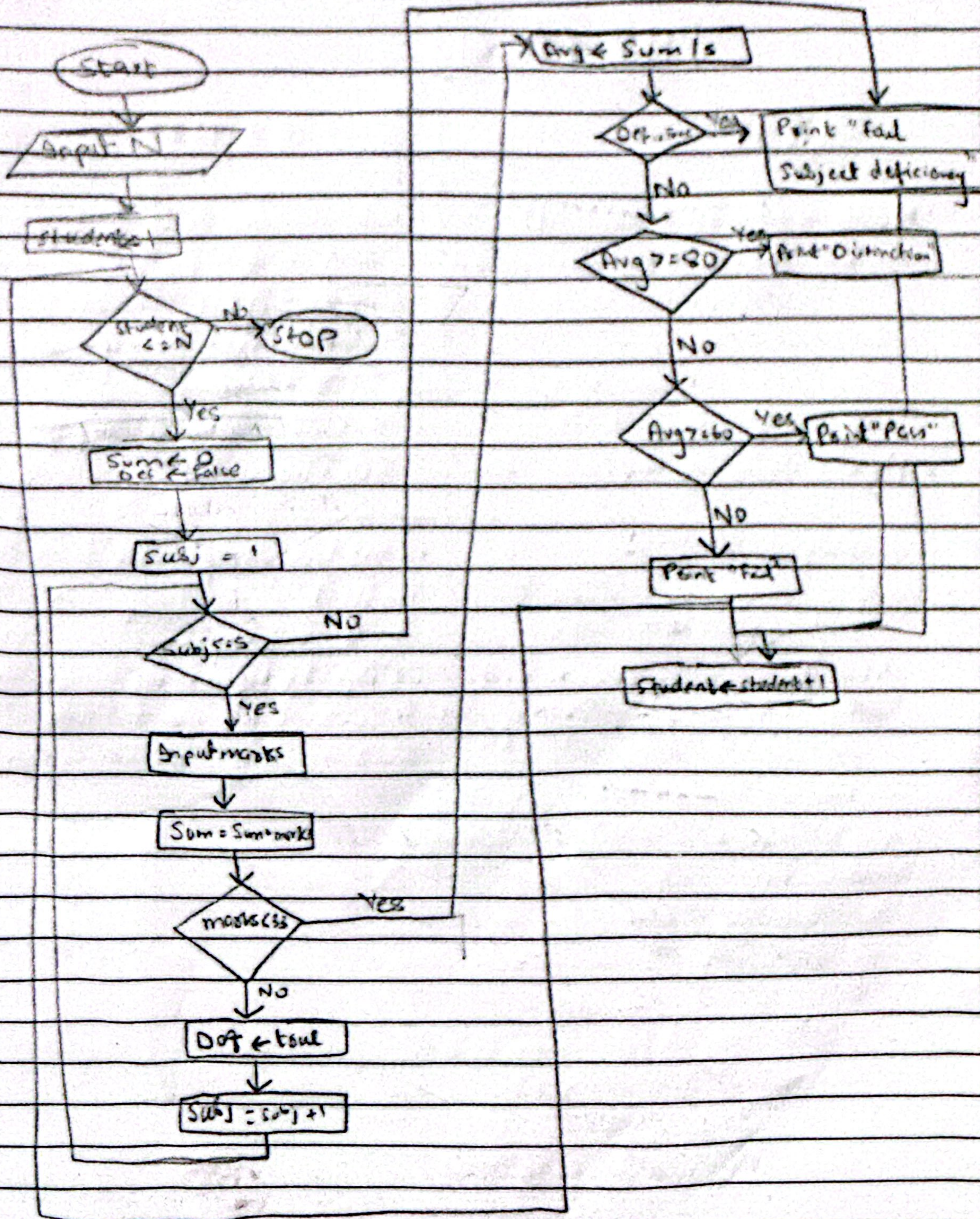
Print ("Fail")

END IF

END IF

END IF

ENDFOR



Dated: _____

PAC Chart:-

INPUT

- Number of students (N)
- Marks for 5 subjects (marks)

OUTPUT

- For each student
 - student number
 - Average marks (Average)
 - Result category:
"Distinction", "Pass", "Fail",
or "Fail - Subject deficiency"

PROCESSING REQUIRED

- Read N
- Loop for each student (1 to N)
 - Reset running sum (Sum = 0) and deficiency flag (Def = "False")
 - Loop 5 times (1 to 5)
 - Input marks
 - Add marks to Sum
 - If marks < 23, set Def = "True"
 - Compute Average = Sum / 5
 - If Def = "True", result is "Fail Subject deficiency"
 - Else if Average >= 80, result is "Distinction".
 - Else if Average >= 60, result is "Pass".
 - Else result is "Fail".
 - Print student's Average and Result

SOLUTION ALTERNATIVES

- Can be implemented using nested FOR loops or WHILE loops.
- Subject deficiency override must take precedence over the calculated average.

Dated: _____

GPO chart

Input	Process	Output
• N Total students • marks (Mark for each subject, = per student)	1) Input 2) For each student (1 to N) a) Initialise Sum = 0, Def = "True" b) for each Subj (1 to 5) i. input marks ii. Sum = Sum + marks iii. If marks < 33, Def = "True" c) Calculate Average: Sum / 5. d) Determine Result • If Def == "True" → "Fail" • "Subject deficiency" • Else if Average >= 80 → "Distinction" • Else if Average >= 60 → "Pass" • Else → "Fail" e. Output student summary	• Student number • Average • Result status